

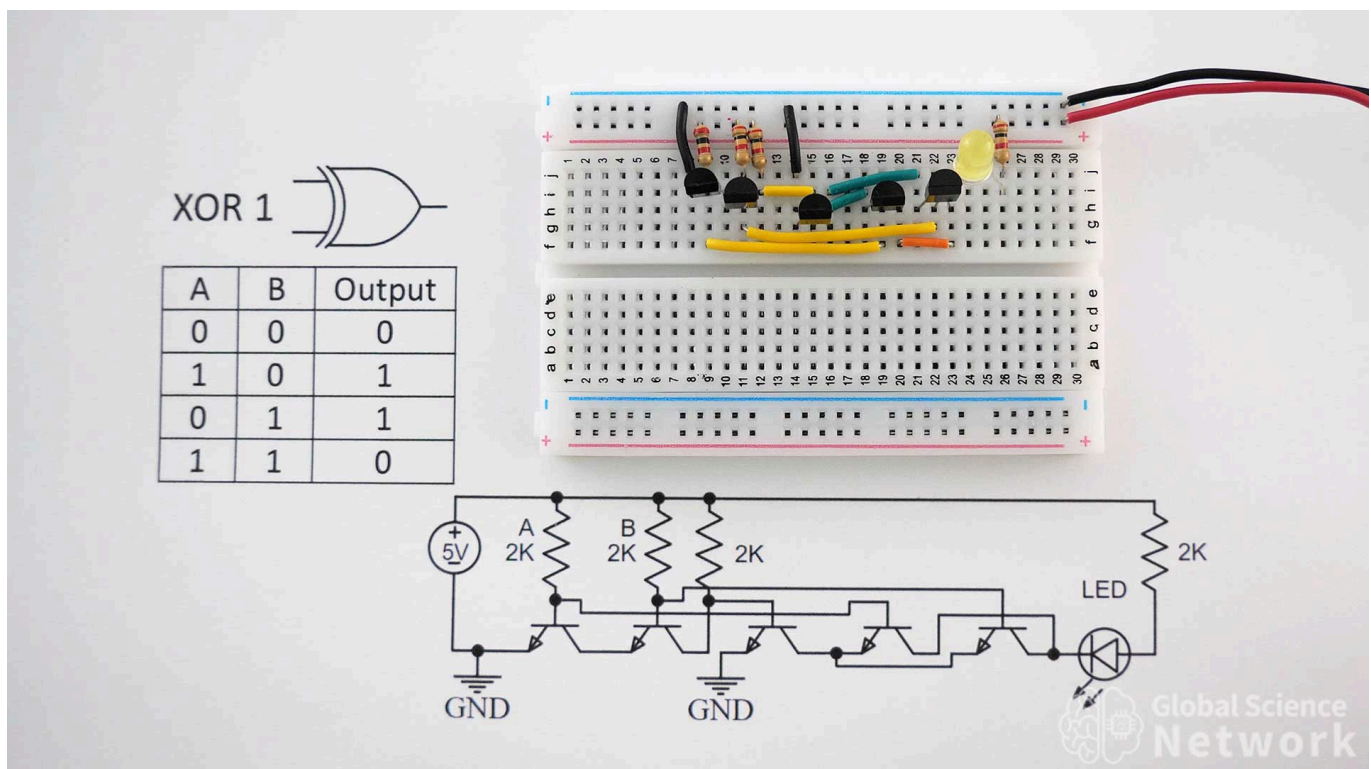
XOR Gate, Exclusive OR Gate



Cody Wabiszewski
July 2, 2024

The XOR gate is also called the exclusive OR gate. In this article, I will show 4 different ways it can be built on a breadboard using individual transistors. This gate is similar to the OR gate but when both inputs are on the output is off. That is why is the exclusive OR gate as in only for the true OR cases. In the simplest design, only 5 transistors are needed. However in order to send an output 6 transistors will be needed. The XOR gate can be built with NAND gates but 8 transistors are needed. It can also be built with NOR gates and 10 transistors would be needed.

XOR Gate 1



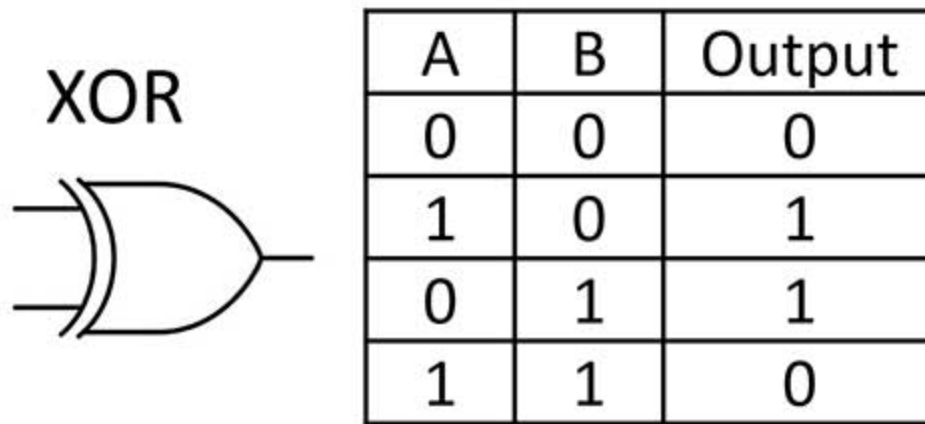
The photo above shows a simple XOR gate that uses 5 transistors. In the layout inputs A and B are both on which is why the LED is off. The two

resistors on the left are the two inputs. Different color wires help to see the connections. If there is any ambiguity in the placement of wires the circuit diagram can be referenced. The gate design is a **NAND gate** on the left two transistors, a switch for the middle transistors, and an OR-like gate for the last two transistors. Looking at the configuration in the photo the current from the far right resistor can not reach the ground to the left so the LED is off. This is the case because all of the current from the first resistor from the left is going to the first ground. When this happens the switch is off because there is not enough voltage going into the base of the third transistor.

When one input is on current is able to flow from the far right transistor to the second ground. Finally when both inputs are off the output is off because the current does flow into the base of the OR gate transistors making it so the current can flow from the far right resistor into the second ground.

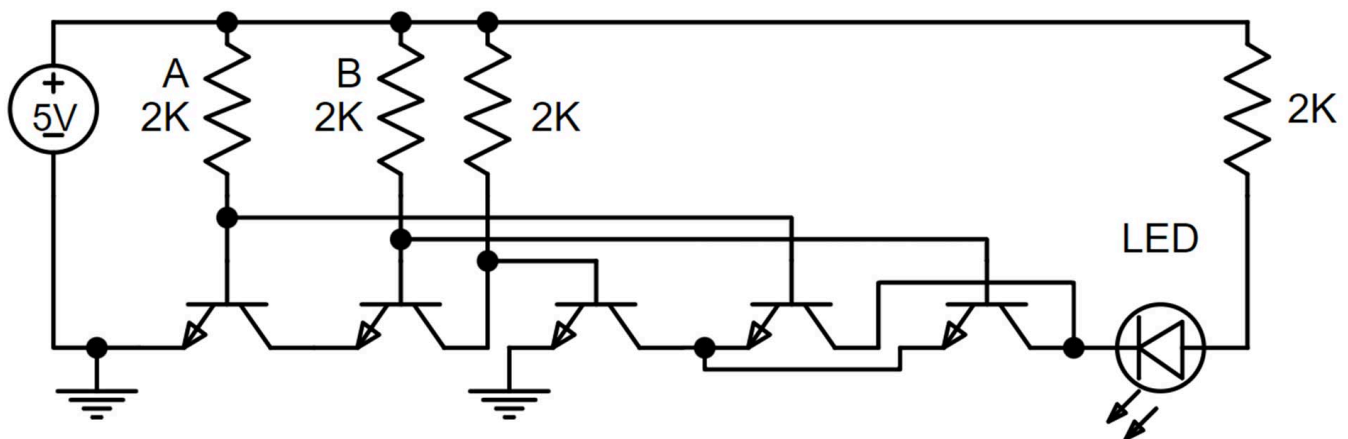
The video above shows how to build all types of logic gates. At **31:48** how to build the Exclusive OR gate is discussed at length. It is common to use XOR gates in **half adders**, **full adders**, and for other computations so this is an important gate to understand.

XOR Gate Truth Table



The XOR gate symbol and truth table are shown above. This table makes it clear that when both inputs are on or off the output is off. When only one input is on the output is on. I will show several different ways to make an XOR gate but the truth table will be the same for all of the cases.

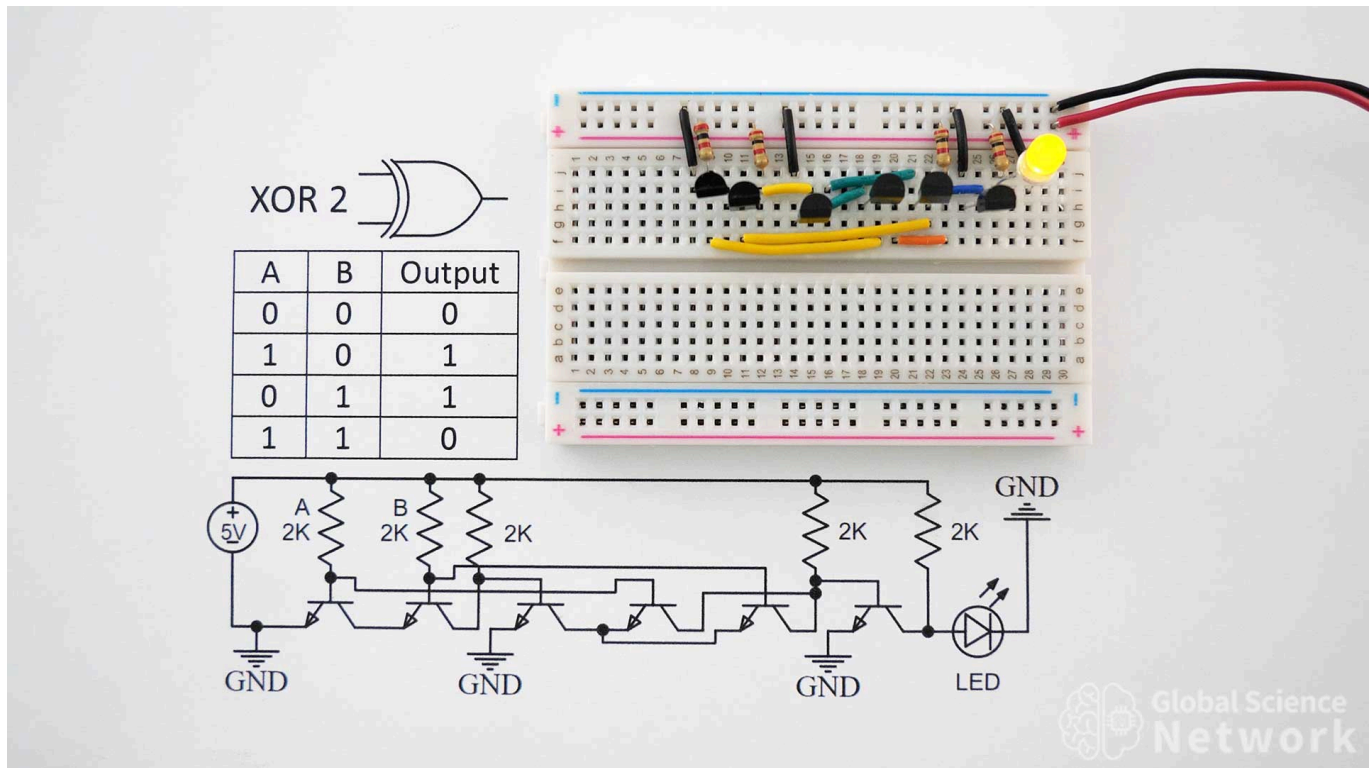
XOR 1 Circuit Diagram



This is the circuit diagram for the first case that uses 5 transistors. Inputs A and B are on when the first two resistors are in place. Now the direction of the LED is configured so current is flowing into the circuit. This means this

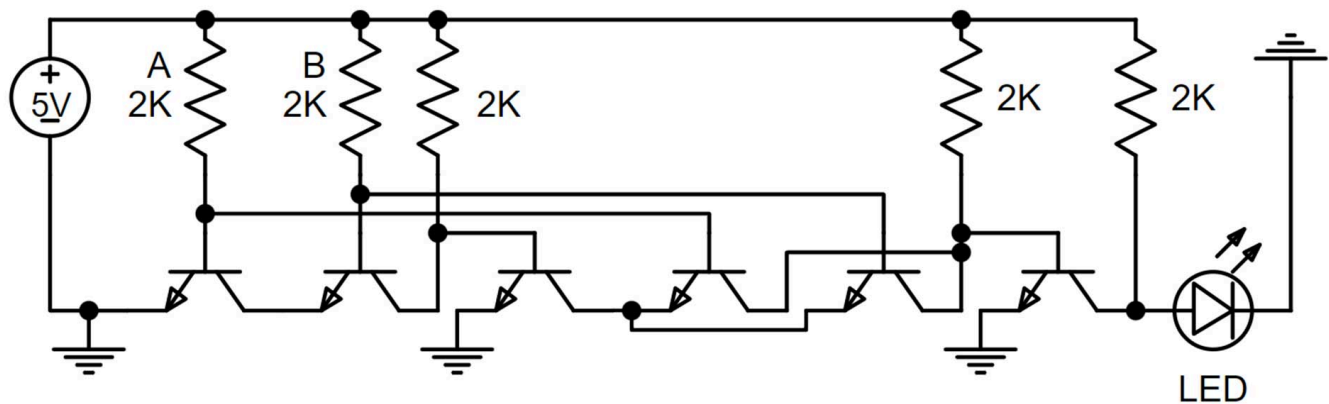
layout will work as a stand-alone logic gate or as a final logic gate. It will not work to send the signal elsewhere.

XOR Gate 2



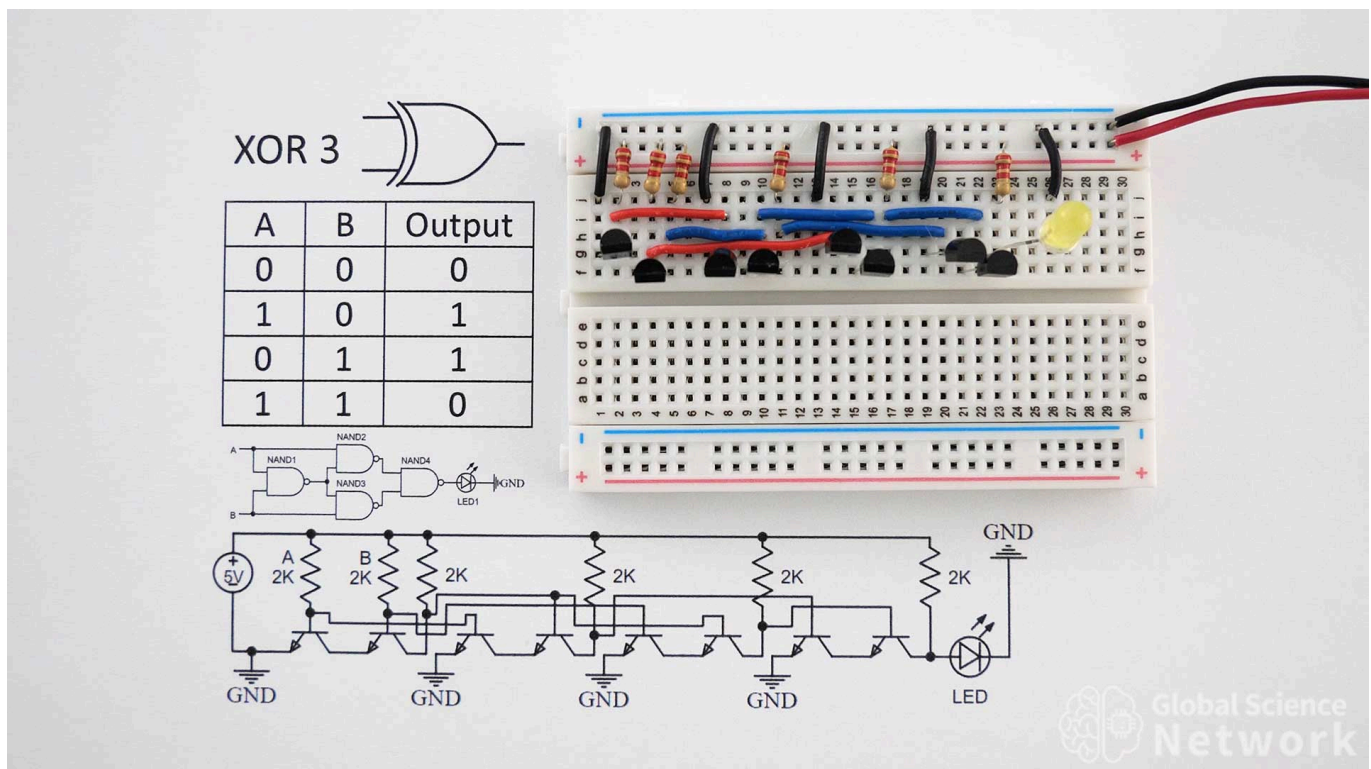
XOR Gate 2 is very similar to the first XOR gate but adds one more transistor on the right-hand side of the circuit. This makes it so that the current can be sent as an output. I would recommend building the XOR gate this way as it works in all cases and only requires 6 transistors. Note that it takes 6 transistors and will work in all cases.

XOR 2 Circuit Diagram



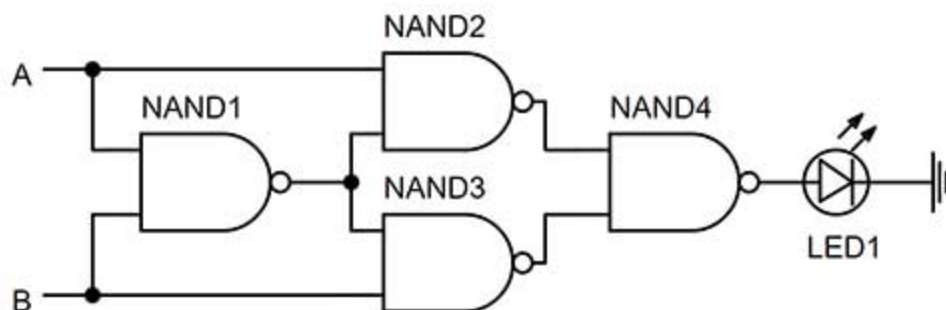
The circuit diagram for XOR gate 2 is almost the same as the first XOR gate. Now there is an additional transistor that is somewhat acting as an inverter. However, now the LED direction is flipped and the current is flowing out of the circuit. This is good as the output can be sent elsewhere. Typically it is sent to other **logic gates** to complete more computations.

XOR Gate Built with NAND Gates

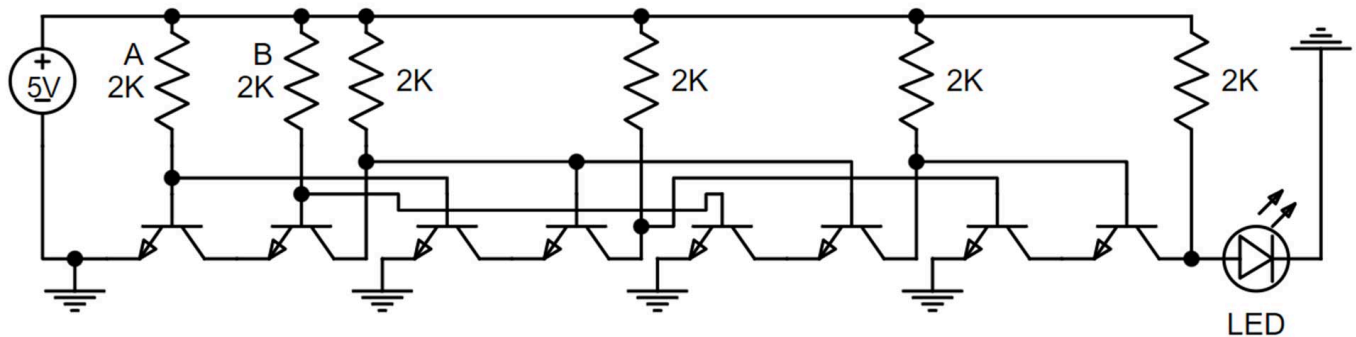


XOR gate 3 is built using 4 **NAND gates**. Each NAND gate requires 2 transistors so a total of 8 transistors is needed to build the gate in this configuration. Input A and input B are the first two resistors on the left-hand side of the circuit. Right now both inputs are on and the output is off, which is expected. To turn the inputs off the input resistors need to be removed. The resistor values are 2K and the circuit is powered with a 5-volt battery pack.

XOR Gate Built with NAND Gates Circuit Diagrams

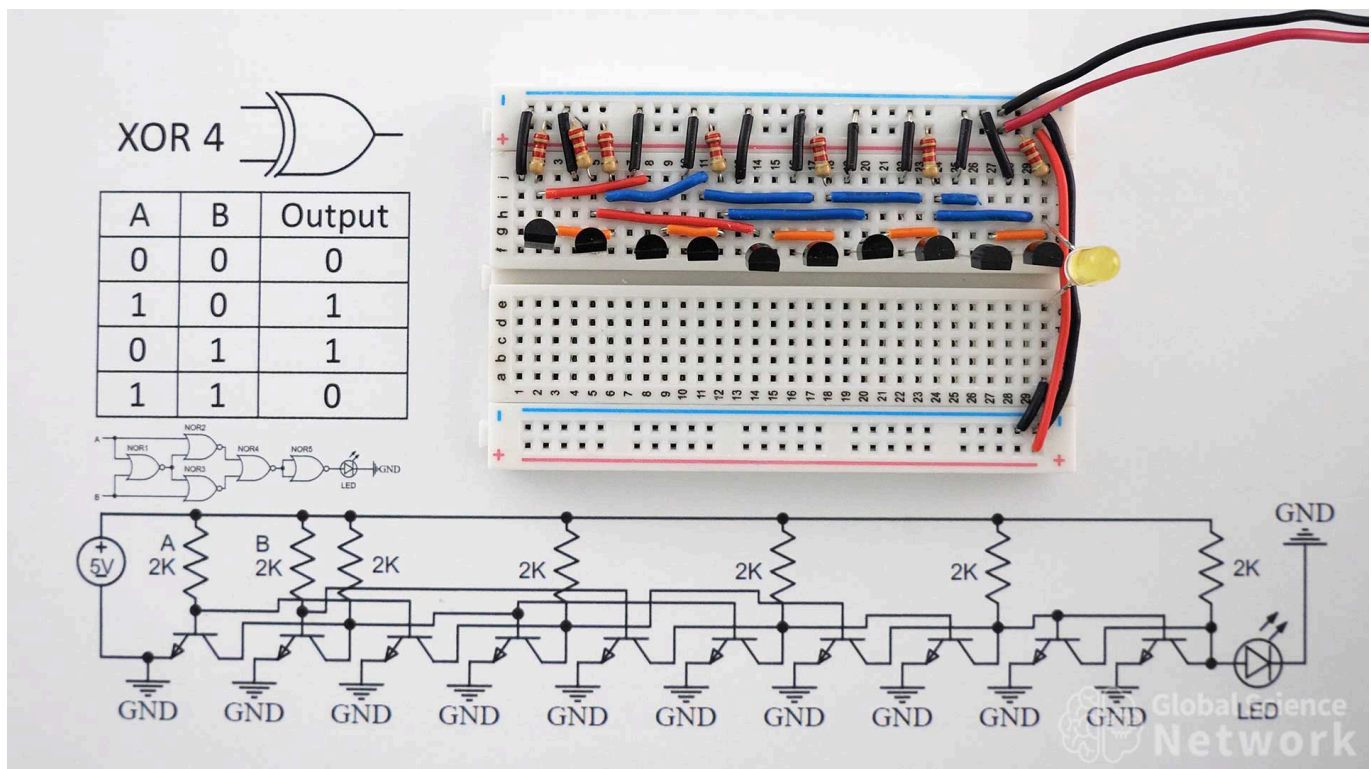


The gate-level circuit diagram is shown above. This shows how the inputs and output of each NAND gate need to be connected to make the NOR gate. At the transistor level, you get more detail including seeing the ground connections. Many times circuit designers are just using integrated circuits for the logic gates and the high-level diagram is sufficient.



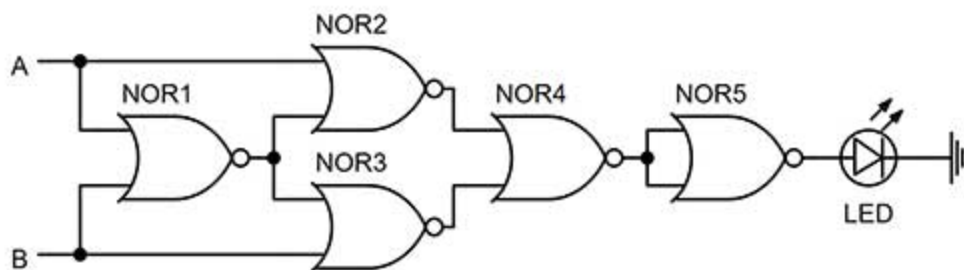
The circuit level diagram above shows component level details of this configuration. It is often easier to see how wires are to be connected by looking at the circuit diagram rather than the circuit on the breadboard. This shows the connections to the base, emitter, and collector of each NPN transistor. The transistors can be model numbers 2N2222 or 2N3904, which are both BJT transistors. All the resistors are 2K but values from 330 ohms to 2.2K would work.

XOR Gate Built with NOR Gates

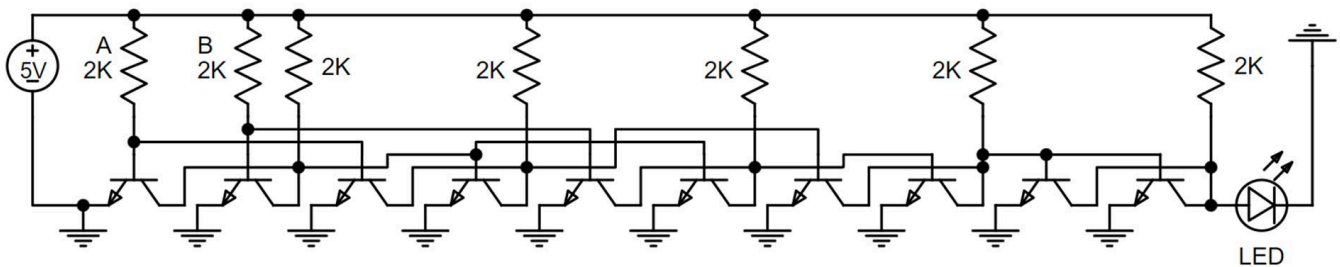


Exclusive OR gate 4 is built using 5 **NOR gates**. Each NOR gate requires 2 transistors so a total of 10 transistors is needed. For this reason, this is not the simplest NOR gate to implement. In most cases, it will not be used unless the computer or circuit architecture is based on the NOR gate. One example where this was the case was the **Apollo Guidance Computer**. There were two versions of the computer, the first used 4,100 NOR gates on separate integrated circuits, and a later version used 5,600 NOR gates.

XOR Gate Built with NOR Gates Circuit Diagrams

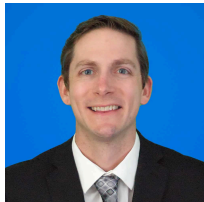


The gate level circuit diagram using NOR gates is shown above. This shows where inputs A and B should be connected as well as where the output of each gate should go. How each NOR gate is made is not detailed and each NOR gate could actually be made from four NAND gates. However, using the two transistor NAND gate is likely the best option.



Above is the component level circuit diagram of NOR Gate 4. The circuit requires 10 NPN transistors, 7 2K resistors, and an LED. The circuit is powered with a 5 Volt power supply.

The video above shows XOR gates used in a 4-bit computer. XOR gates are used in the ALU for the subtract feature and in two of the full adders.



[Cody Wabiszewski](#)

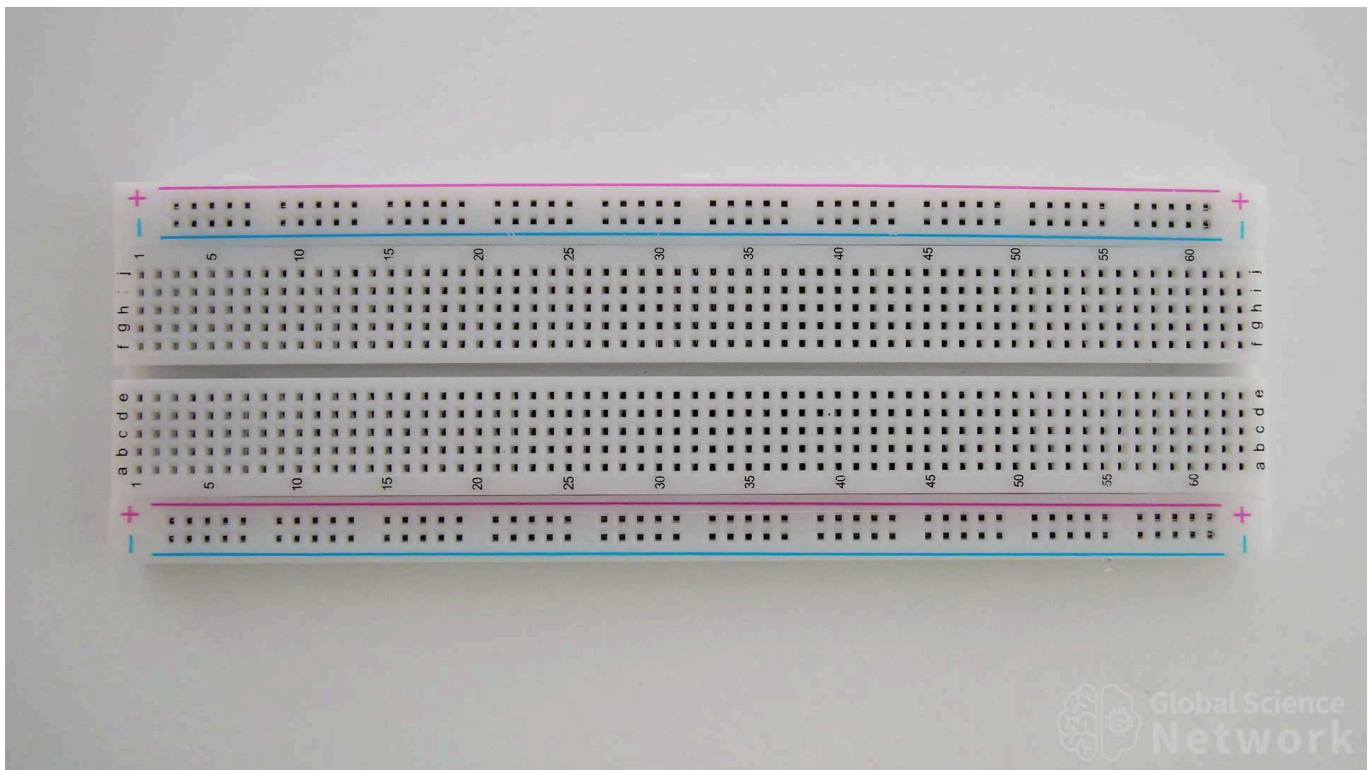
Cody started the Global Science Network with the idea people should be focusing more time, energy, and resources on useful projects. He has a bachelor's degree in aerospace engineering and a master's degree in mechanical engineering. Cody has worked for the US federal government, a university, a large corporation, small businesses, and for himself. He has done human brain computer interface research and is currently working towards creating non-biological human consciousness.

Breadboard, Components, Tools, and Power Supply

Breadboards are used to build and test electrical circuits. These are used by hobbyists and professional electrical engineers. Using a breadboard allows for electrical connections to be made without using solder which is why they

are also called solderless breadboards. Not having to make a permanent connection with solder makes it so the placement of wires and discrete components such as resistors, LEDs, and transistors can be easily changed.

A breadboard is typically used in the prototyping stage of a circuit design. This makes it so that mistakes can be made without having to build an entirely new printed circuit board (PCB). Different designs can be tested quickly with almost instant results. Sometimes it is easier to design circuits in a circuit simulator. However, at some point in the circuit design, it should be tested with actual components to verify it works as expected. Once a final design is made on a breadboard it can then be made on a PCB which makes the circuit smaller, and easier to mass produce.

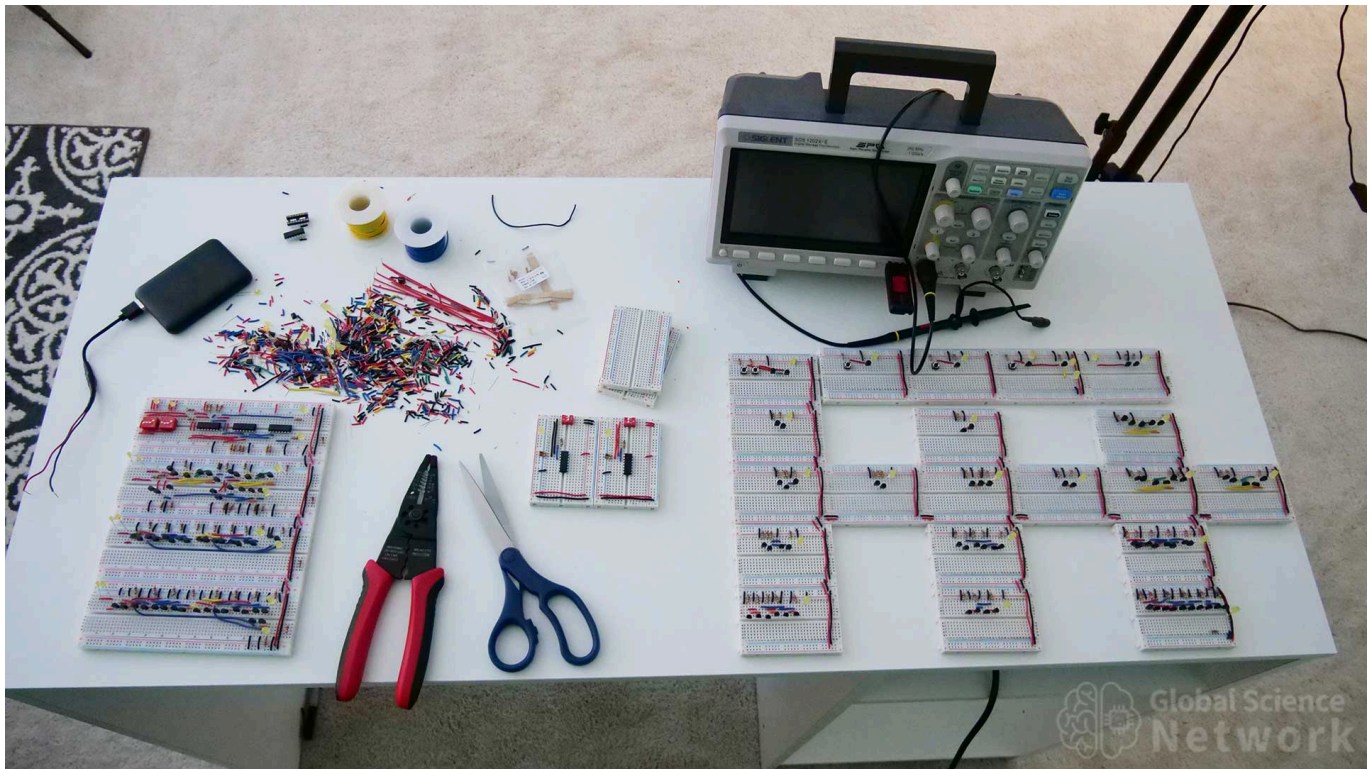


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There are several different size breadboards but each one is built in a similar way. Power is supplied to the power rails which run along the outside of the breadboard. The power rails are horizontal rows. On the inside, there is a grid of holes where the rows are labeled a-j and the columns are numbered 1-60. The holes in the power rail rows are electrically connected but the rows on

the inside are not electrically connected. Vertical columns are electrically connected between rows a-e and rows f-j.

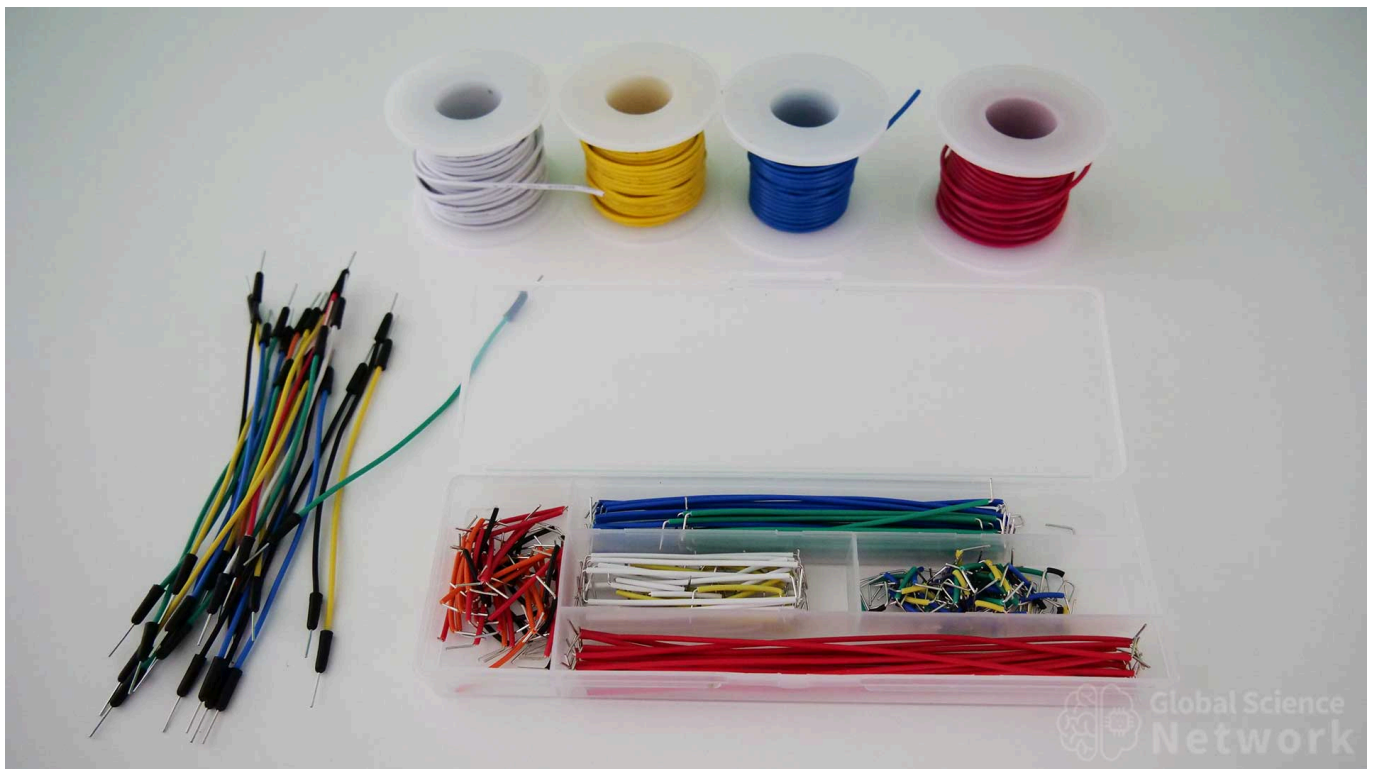
The power rails are labeled + and – for where the positive and negative terminals of the power supply should be connected. Multiple breadboards can be connected together by sliding the tabs of one breadboard into the grooves of another. A breadboard can also be secured to a flat surface as the bottom has a sticky layer that can be exposed by removing the paper backing.



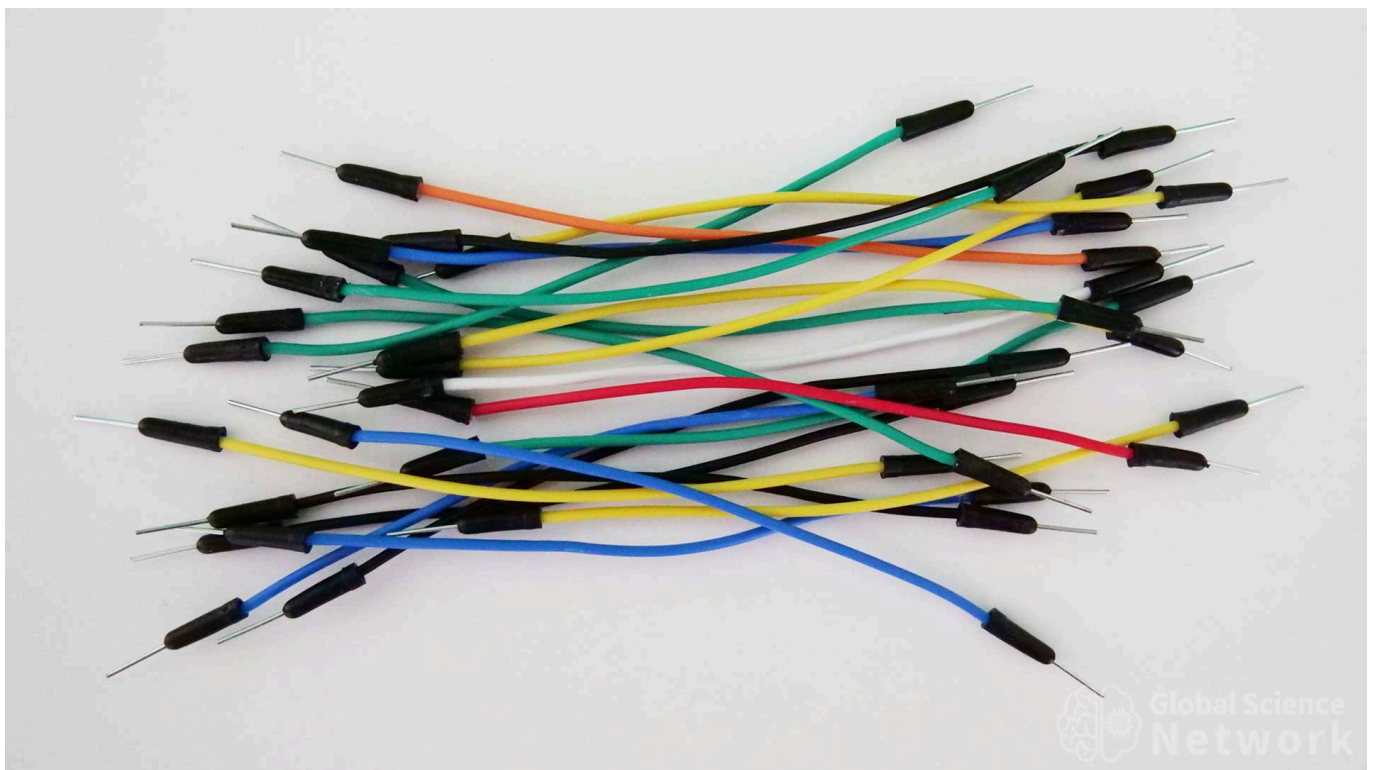
In the photo above I am working on multiple breadboard projects involving digital logic gates. An oscilloscope is used to measure voltages within the circuits. On the desk, there is a large number of cut wire scraps as each wire is custom cut to length. Both full-size and half-size breadboards are being used.

Breadboard Components

Wire

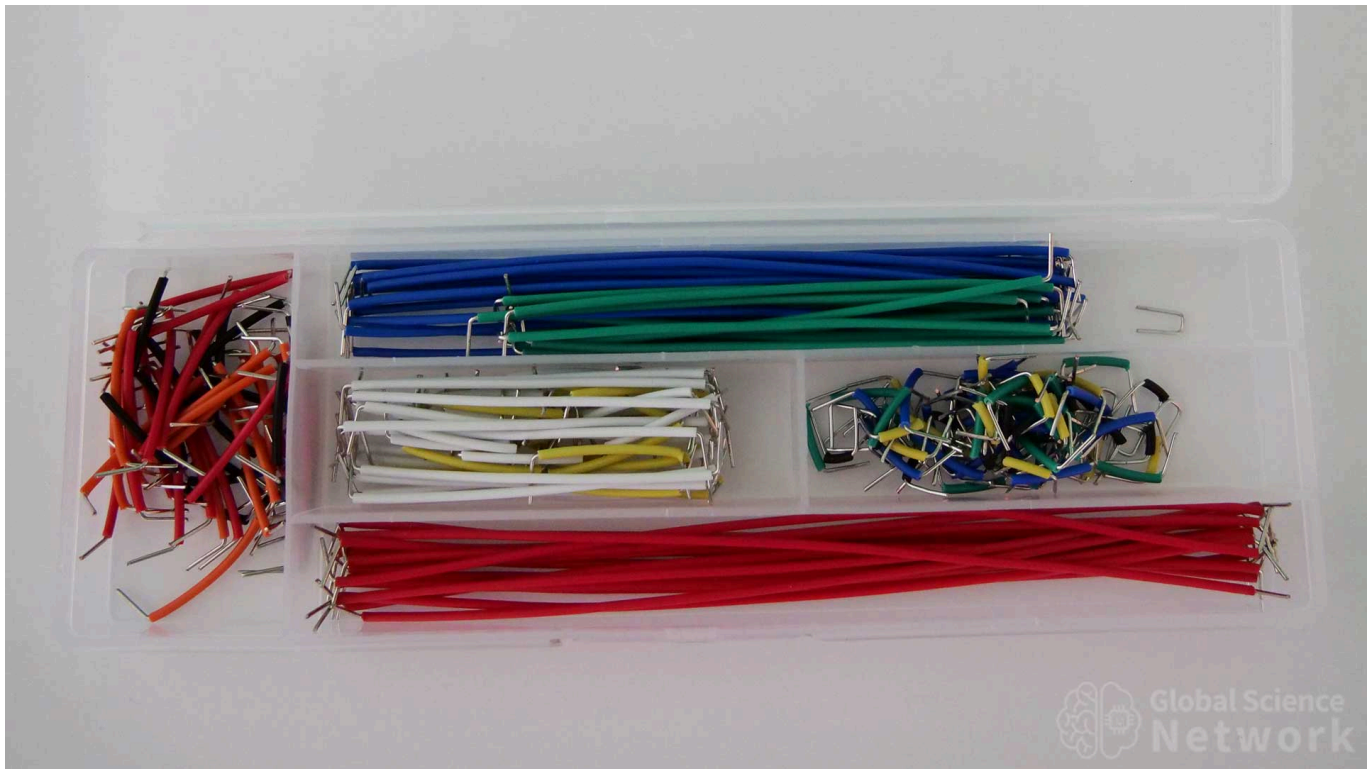


There are three different types of wires used for breadboarding. These are jumper wires, spools for wire, and pre-cut wire. Each will be discussed in detail below.



Flexible jumper wires are the quickest and easiest way to make a wire connection on a breadboard. These are made with multistrand wire with a

PVC coating. Multistrand cable is flexible but does not fit well into breadboard holes. For this reason, there are metal pins on the end that fit properly into the holes of the breadboard. The disadvantage of using jumper wires is the extra length of wire makes it less clear where each wire is going to and from. This is especially true when lots of wired connections are made. Jumper cables are great for quick testing but are not to build circuits to take pictures of and show circuit designs to others.



Precut wires can be nice because the wires are cut and bent to specific lengths. This does help speed up the process when breadboarding. The problem is that the wires are color coated based on length. Often times when building circuits it is nice to color code the wires based on the connection type. For example black going to the ground, red going to the positive rail, blue wires sending data, etc. I do use precut wire when the wire I need is shorter than the precut wire. This way I can cut it down and still use the color I want. It saves some time as one end is stripped and bent already.



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The best wire to use for breadboards comes on spools. It is a 22 gauge solid core wire. Each spool has over 25 feet of wire. Common wire colors are red, black, yellow, blue, white, and green. The disadvantage of the wire spools is that each wire needs to be cut to length and stripped on both sides to expose the wire. When this is done properly it makes for a very nice-looking circuit where it is easy to see where the wires are going to and from. So it is often worth the extra time. Each wire should be cut so that it lays flat and the stripped ends fit securely in the holes.

Resistors



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Resistors are one of the most common components used when breadboarding. Each resistor is marked with color-coated bands to display its value. Common resistor values used for breadboards range from 1 ohm to 1 mega ohm which is 1 million ohms. Often times the exact value of the resistor is not needed so one with a slightly higher value should be used. For example, if your calculations show you need a 945-ohm resistor, using a 1K resistor will work in most cases.

Resistors come with long wires on each end that need to be cut and bent to fit properly in the breadboard. The full length can be used but in most cases, this will make the resistor sit high above the breadboard. Having resistors lie flat makes the breadboard look more organized and is easier to see where connections are being made.

LEDs

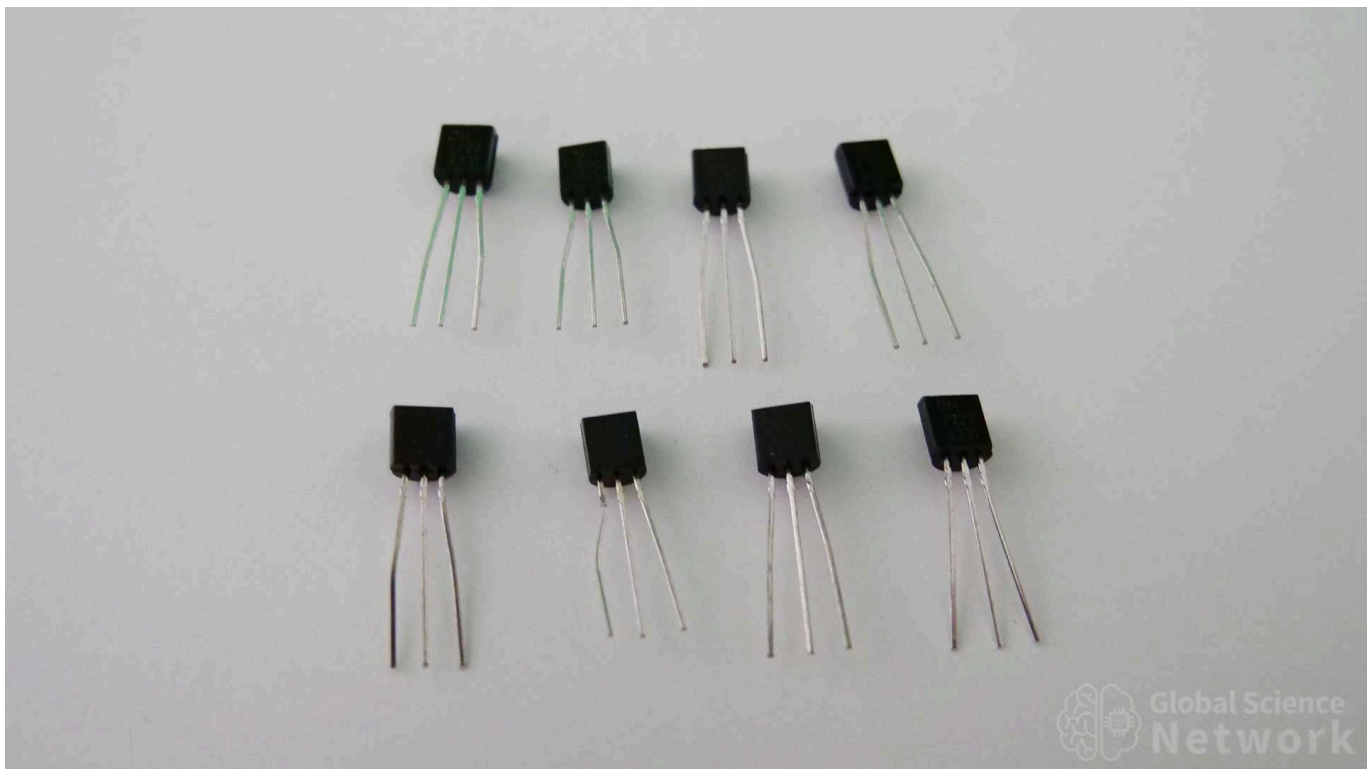


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LEDs are often used in breadboards to show that the power is on. Using an oscilloscope or multimeter can provide the exact power value. However, often times it is advantageous to just see that power is on. Also, the brightness of the LED gives insight into how much current is being applied across the LED. When using LEDs it is important to have a **current-limiting resistor** or the LED can be damaged.

Most LEDs have a voltage drop ranging from 1.8 volts to 3.2 volts. This is different from standard diodes which have a voltage drop of around .6 volts. The common input voltage for breadboards ranges from 3 volts to 9 volts. If the resistor is larger than 350 ohms it should limit the current to less than 20 milliamps. The max current rating for LEDs is typically 20-25 milliamps. Common colors for LEDs are red, yellow, white, blue, and green.

Transistors



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BJT transistors have three pins that are designed to work with breadboards. The three pins are the emitter, base, and collector. Which pin is which will vary depending on if it is an NPN or PNP transistor. The most common transistor used with breadboards is the 2N2222 NPN transistor. This is an amplifying transistor and also works as a switching transistor. I have used these to build [digital logic gates](#). Like LEDs, transistors should also have a current-limiting resistor to prevent damage.



Transistors often come in large sets where there are many types of NPN and PNP transistors. The most common NPN transistors are the 2N2222 and the 2N3904. For PNP transistors the most common are the 2N2907 which is equivalent to the 2N2222 and the 2N3906 which is equivalent to the 2N3904. Each transistor does have different specifications so using the proper model number is important.

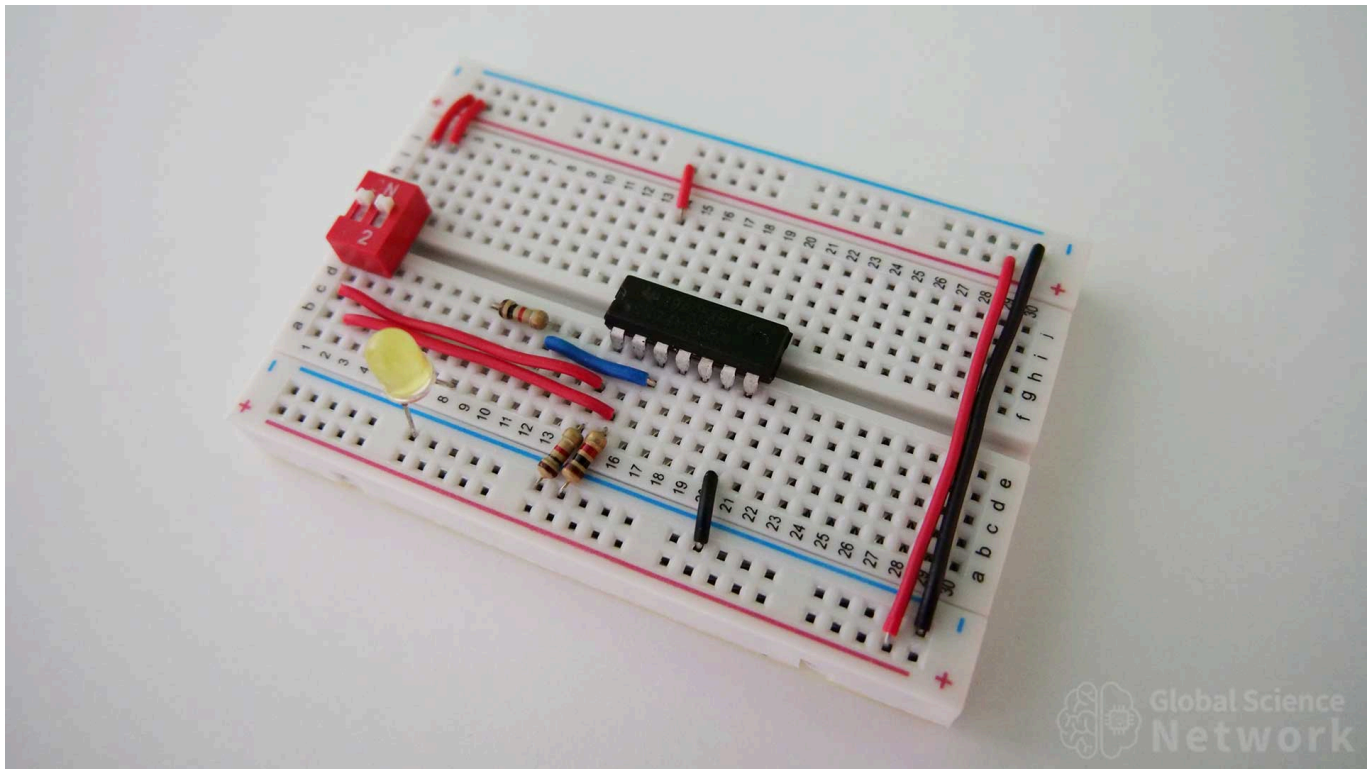
Capacitors



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The three types of capacitors used with breadboards are ceramic capacitors, aluminum electrolytic capacitors, and metalized polypropylene capacitors which are also called film capacitors. In the photo above is a set of ceramic capacitors. These have a capacitive range from $0.1\ \mu\text{F}$ to $10\ \mu\text{F}$. Ceramic capacitors are non-polarized meaning the orientation of the capacitor is not important. Electrolytic capacitors are polarized meaning one terminal must be connected to the positive voltage and the other side must be connected to the ground side. If this is done in reverse the capacitor will quickly heat up and break, often resulting in a popping noise and the smell of melting capacitor material. Most film capacitors are not polarized. Breadboard capacitors can range in size from $10\ \text{pF}$ to $10,000\ \mu\text{F}$.

Integrated Circuits



Breadboards are also designed to work with integrated circuits. An entire microprocessor can be placed on an integrated circuit so ICs drastically increase the potential of what can be done on a breadboard. Common integrated circuits to use on breadboards include logic gates, 555 timers, op-amps, shift registers, EPROM, FPGAs, gyroscopes, and microprocessors.

BreadBoard Tools

Wire Stripper



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Wire strippers are needed to cut wires and remove the wire insulation on the end of the wire. Breadboard wires are 22 gauge so the stripper needs to have a slot to strip 22 gauge wire insulation. The wire stripper above can cut and strip wire from 10 gauge to 22 gauge. It also has a crimper to make crimp connections. When cutting breadboard wire to exact lengths some type of cutter and wire stripper is needed.

Wire Cutter



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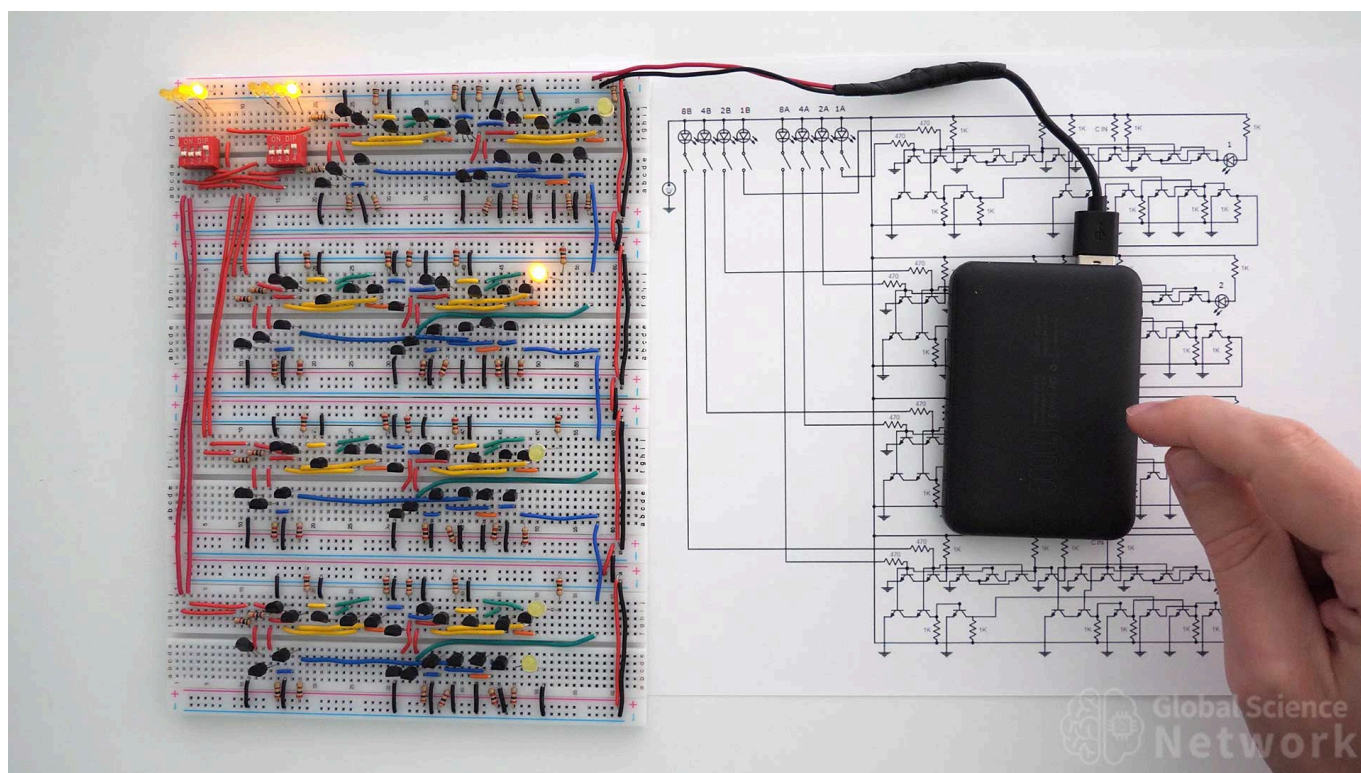
Scissors or a wire cutter make cutting wire much easier. The wire stripper tool does have a cutter but only on the back of the jaws. Having a tool with a cutter on the front jaws makes it easier to mark where the wire needs to be cut. Resistor wire is very thin and is easy to cut with scissors. I often cut multiple resistor wires down to the proper size with just one cut on each side. A small wire cutter is probably best but scissors do work if that is what is available.

Breadboard Power Supply

A breadboard can be powered by a variety of power supplies. Batteries used to be the most common way to power a breadboard. Two 1.5-volt batteries in series make a 3-volt power source. Three 1.5-volt batteries in series can be used to supply 4.5 volts. One 9-volt battery works with connector clips and power wires. There are also breadboard power supply modules that can provide 3.3V or 5 volts. The module can receive input power through a 2.1 mm jack plug or USB cable. The disadvantage of the breadboard module is

that it takes up space on the breadboard. Power can also be sent from an Arduino, Raspberry Pi, or microcontroller to the breadboard.

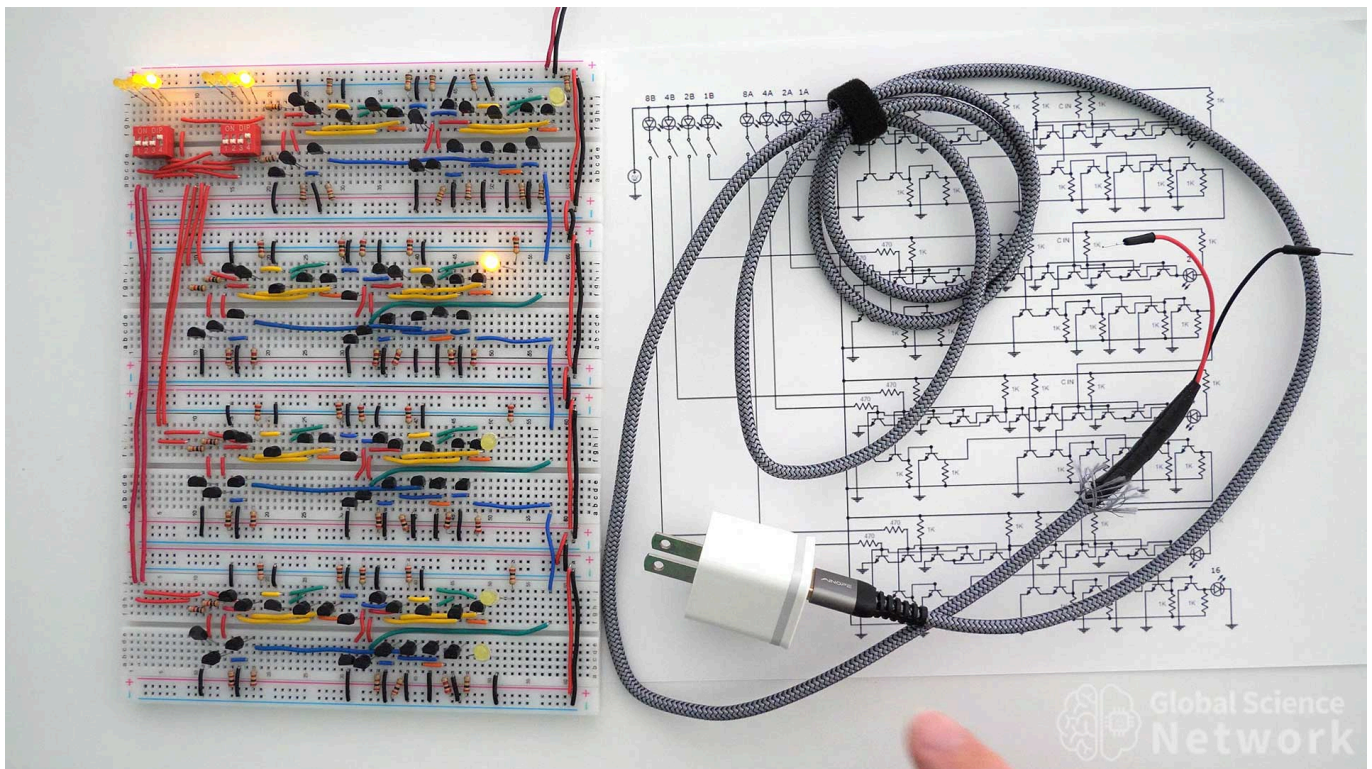
USB Battery Pack



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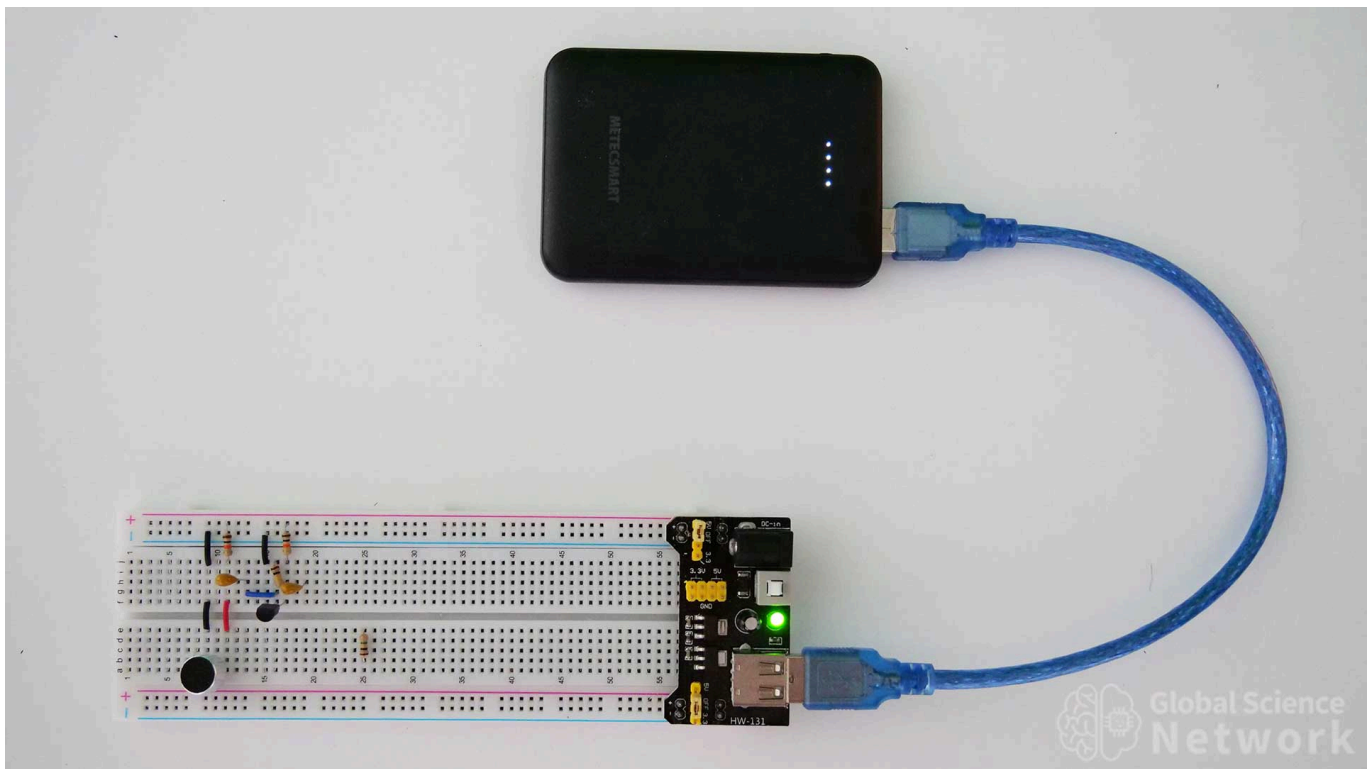
In the photo above I am providing power to a **4-bit calculator** that is built on breadboards using a 5-volt battery pack. I modified a USB power cable so that it could go into the positive and negative terminals of the breadboard. This was done by splicing solid core 22 gauge wire from the stranded cable that was in the original power cable. The only downside to using this battery pack is it does have an auto shut-off feature if the output current is too low. It stops supply power after about 30 seconds when this is the case. Pushing the button on the battery pack will allow the current to flow for another 30 seconds. This is ok when testing circuits because you typically know within 30 seconds if the circuit is working properly. For larger circuits, it will not shut off as the current used is high enough to not shut the power off.

Cell Phone Charger and USB Cable



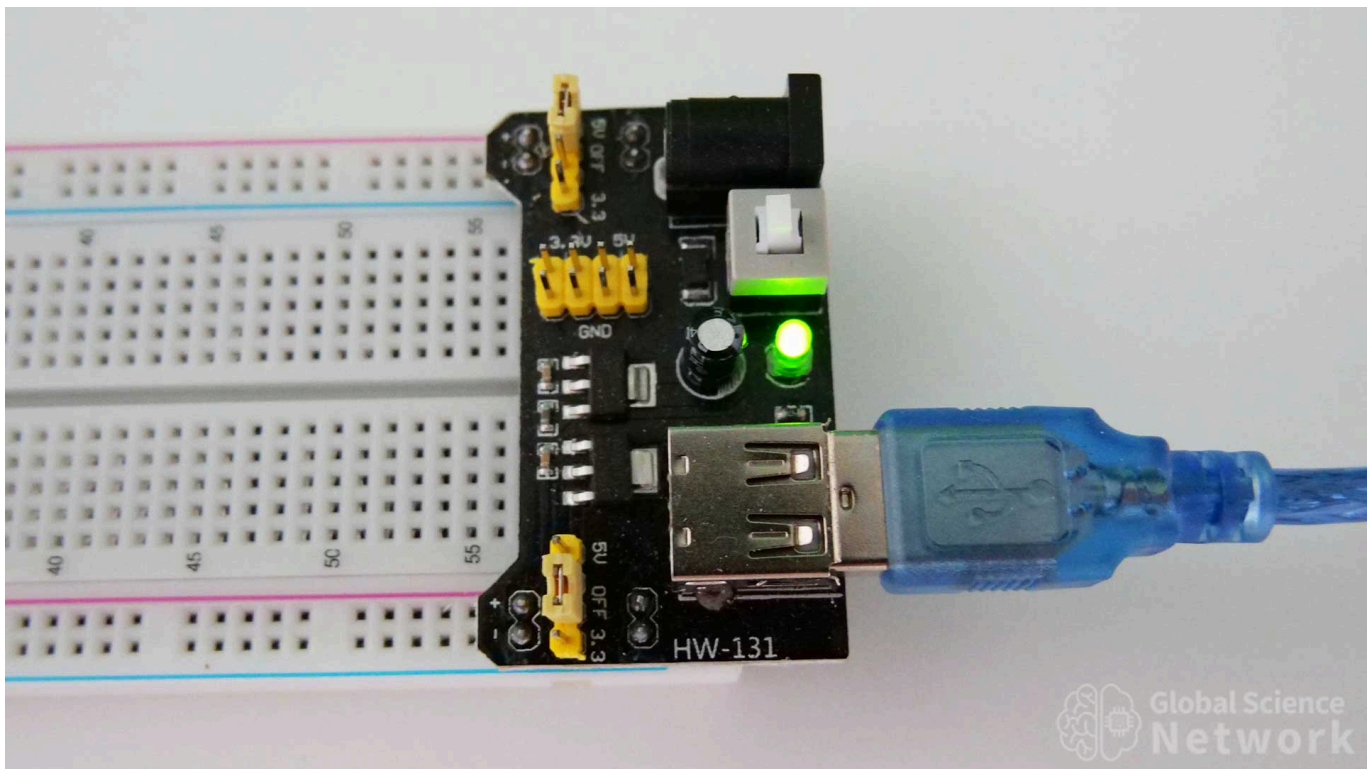
Using a 5-volt cell phone charger is becoming a popular way to power a breadboard. Most cell phone chargers and USB ports from computers provide a 5-volt power source. In the photo above I modified a 6-foot USB cable to have breadboard jumper cable ends. Now, most USB power supplies will work to power the breadboard at 5 volts.

External Circuit With USB Power Supply



The photo above shows how to connect power to a breadboard power supply module using a USB cable. A 5-volt battery pack is supplying the power. The module receives power from the USB cable. Jumpers above each power rail can be moved so that 5 volts, zero volts, or 3.3 volts is supplied to the power rail. The top and bottom power rails can be set to different values. In the photo, the top power rail is set to 5 volts and the bottom power rail is set to zero volts. A green LED on the board indicated the power is being supplied to the board. There is a button that turns the power off to the board and the LED will shut off when power is not being supplied to the breadboard.

Power Supply Module



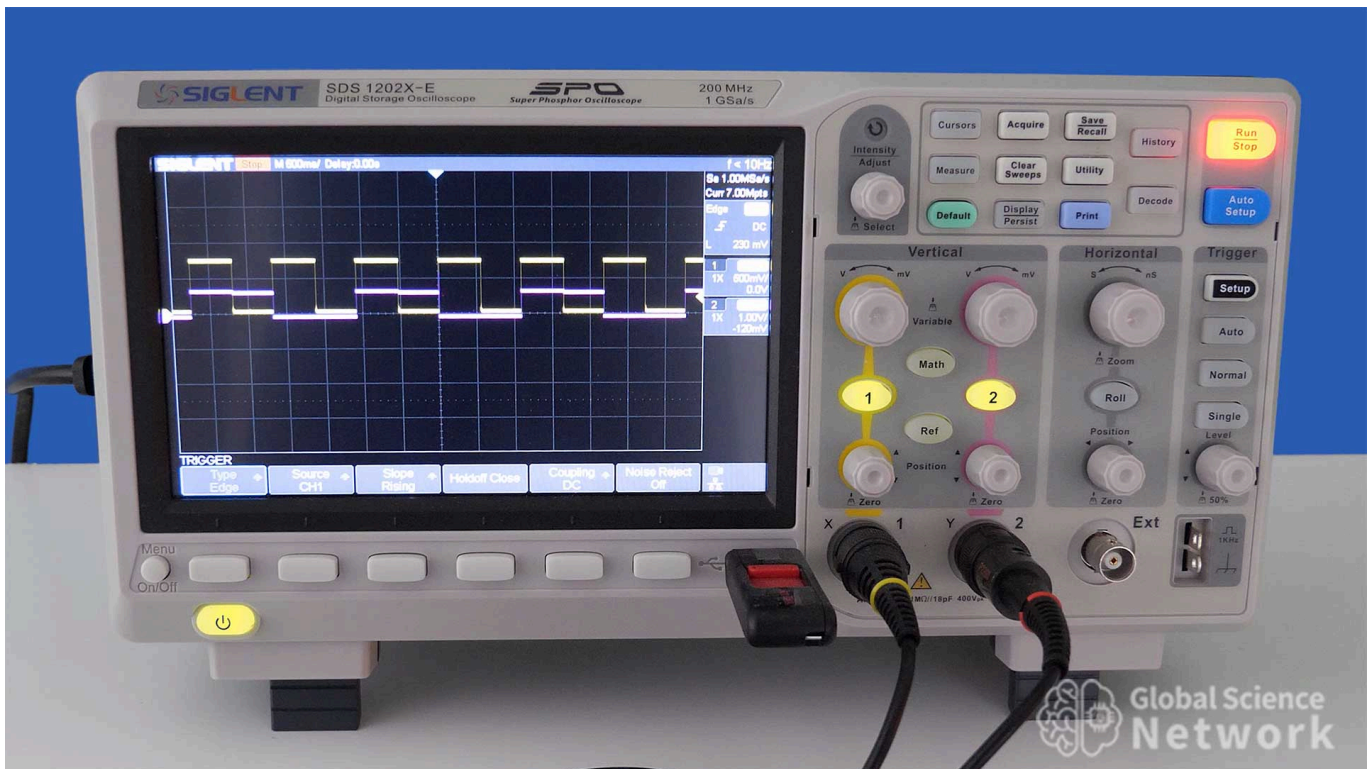
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The close-up view of the power supply modules shows all the features of the board. Input power can come from the jack plug or USB cable. Power going into the power jack needs to be between 6.5 and 9 volts. Power coming from the USB should be around 5 volts. The module is connected to the breadboard with 8 pins. Four pins on each side go into the power rails. This is an ok option to power a breadboard. I personally prefer when power is supplied over two wires. It makes it so there is more room on the breadboard and makes the breadboard less cluttered. When making demonstration videos it is also nice not having people wonder what is going on with the power supply. It is simpler to just show a red wire and a black wire going into the positive and negative slots in the breadboard.



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A multimeter provides a way of measuring the current in the circuit. It is also a way to measure the voltage levels within the circuit. I do use this multimeter to determine the current. For voltage levels though, I prefer to see the output on the oscilloscope as this provides transient information as well.



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Having a digital oscilloscope is really handy when working with circuits. It is often difficult to know what is going on within the circuit without looking at the voltage values at different locations within the circuit. The oscilloscope not only shows the voltage values but shows the values over time so you can see transient conditions within the circuit.

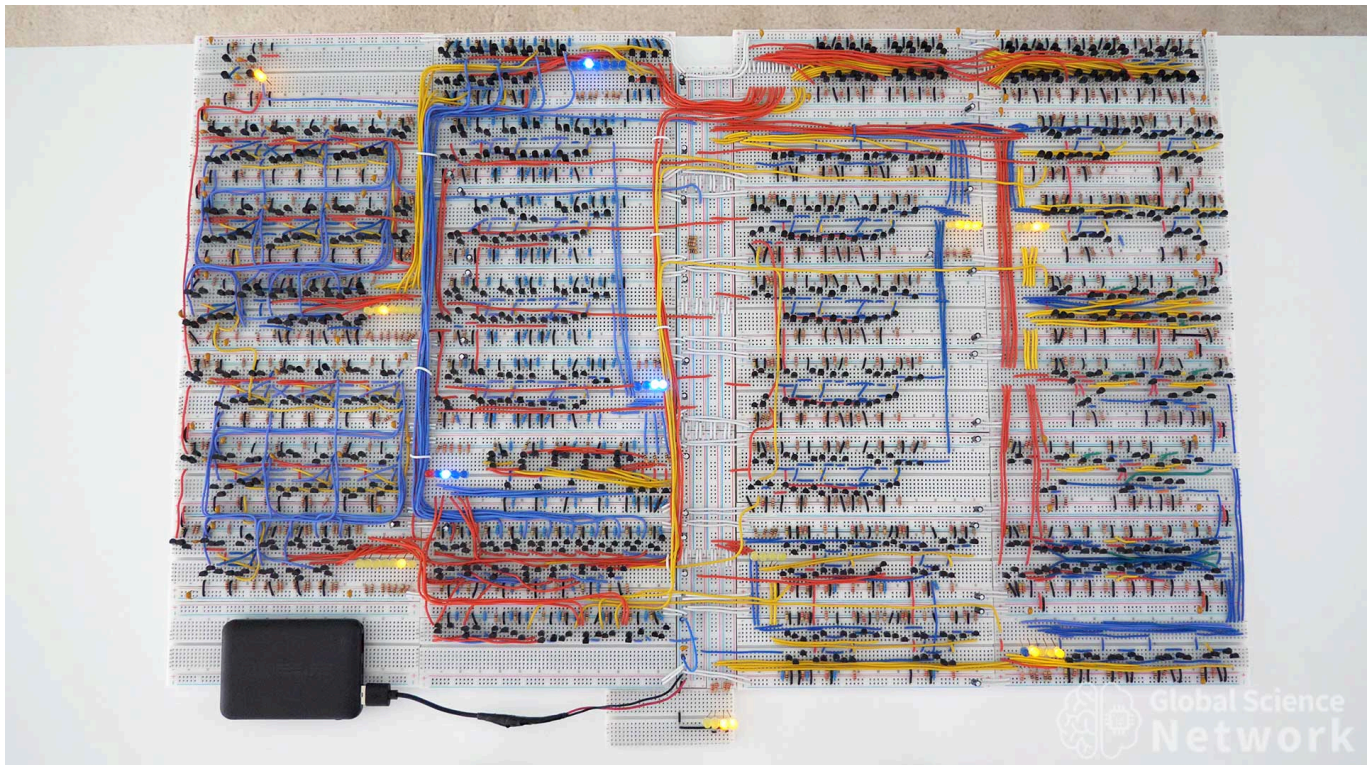


Cody Wabiszewski

Cody started the Global Science Network with the idea people should be focusing more time, energy, and resources on useful projects. He has a bachelor's degree in aerospace engineering and a master's degree in mechanical engineering. Cody has worked for the US federal government, a university, a large corporation, small businesses, and for himself. He has done human brain computer interface research and is currently working towards creating non-biological human consciousness.

4-Bit Computer Built on Breadboards Using Individual Transistors

To learn how computers work it is helpful to build a simple computer. This 4-bit computer is built on 32 **breadboards** and uses 962 NPN BJT transistors. It is powered by a 5-volt rechargeable battery pack and uses about 1 AMP of current. Other components used include resistors LEDs, capacitors, and 22 gauge wires. It is awesome that a functional computer can be built using only these discrete electrical components.

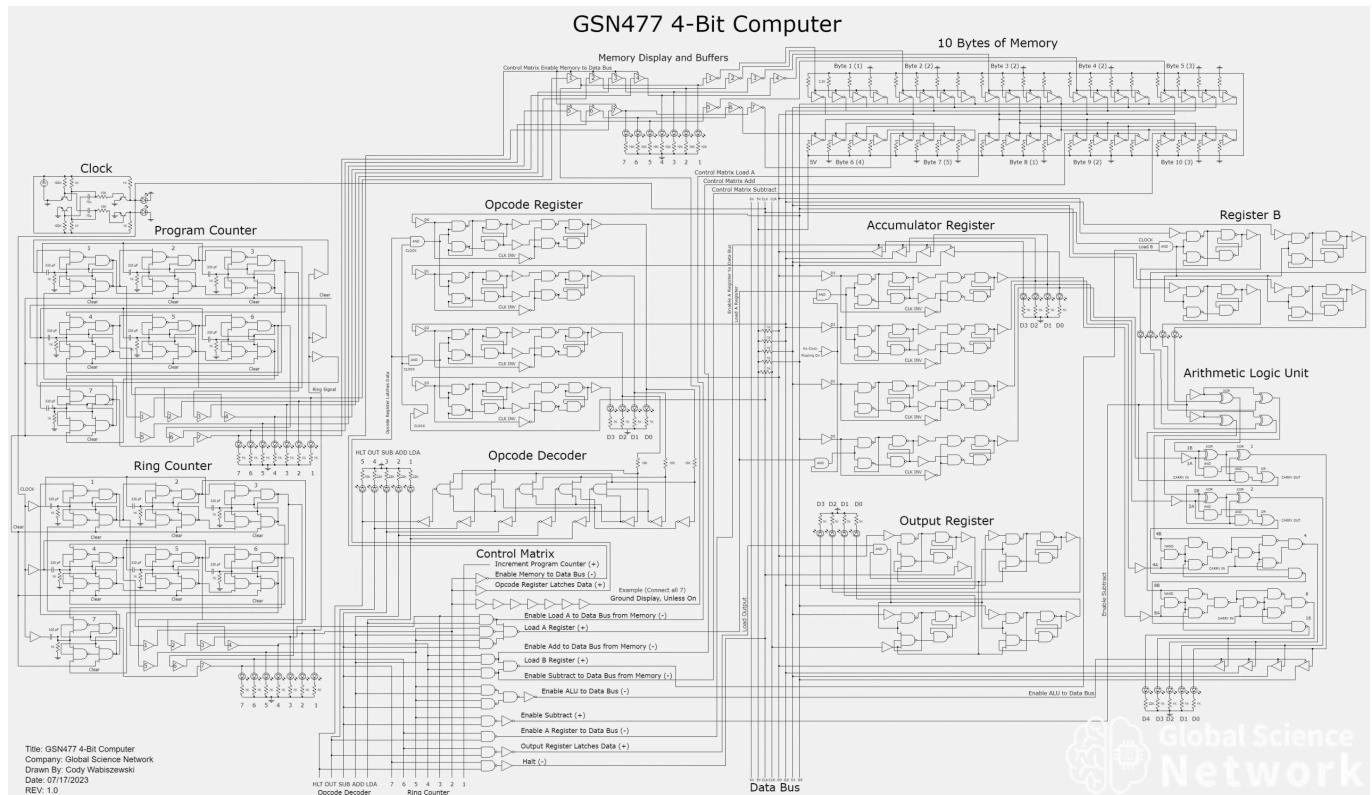


The photo above shows the completed computer. When operating the computer can add or subtract values up to 15. It can hold seven operational commands and store three values to load A, add, or subtract.

The video above describes exactly how the full computer works. I would highly recommend watching the entire video.

When the computer was about half finished I made the video above. The computer is actually functional however the registers, databus, and ALU

signals need to be controlled manually. This is actually a very helpful way to show exactly how information is being sent around the computer and performing mathematical operations.



A complete logic gate-level circuit diagram of the 4-bit computer is shown above. Sections of the computer include the clock, program counter, ring counter, opcode register, opcode decoder, control matrix, data bus, 10 bytes of 4-bit memory, accumulator register, output register, register B, and the arithmetic logic unit which is also called the ALU.

Computer Clock

The computer clock is built with an a-stable multi-vibrator. This circuit produces an even square wave where the output signal is either on or off. It is built with four transistors and two 10 microfarad capacitors. In the video, the clock is completing a full cycle once every 1.75 seconds. The clock rate can be increased significantly higher than this but to see how the computer works this is a good clock rate.

Program Counter

In most computers, the program counter is a binary counter which is shown in the video below. For the 4-bit computer, it was easier to have the program counter be a 7-stage ring counter. This is because if a binary counter was used the memory module would have had to decode the binary number into individual signals. The ring counter already has the individual signals as the default output.

Ring Counter

The 7-stage ring counter was built with four edge-triggered data flip flops. Each trigger is built with a 330 picofarad capacitor and a 1K resistor. Each time the clock completes one full cycle the ring counter increments. The output of the ring counter goes to the control matrix and enables when information will be sent to different parts of the computer.

Opcode Register

Information is sent from the memory module into the opcode register where this value gets latched. Once the data is latched it is continuously being sent into the opcode decoder. The opcode register gets enabled from a control signal coming from the control matrix.

Opcode Decoder

A binary value is continuously being sent from the opcode register into the opcode decoder. Using **inverters** and multiple input **NAND gates** the opcode decoder turns the binary value into a single on output which represents load A, add, subtract, output, and halt commands. These commands get sent into the control matrix to control when circuits in the computer are enabled.

Control Matrix

Signals from the ring counter and opcode decoder are sent to the control matrix. Depending on the values of these signals will determine the output of the control matrix. Circuits in the control matrix are inverters or multiple input NAND gates.

Data Bus

The data bus allows signals to be connected from one circuit to another. Only one circuit should be sending data to the data bus at one time. By default, the data bus has all the data lines on. This is because pull-up resistors are connected from positive 5-volts to each data line. When a circuit connects to the data bus it actually just grounds the data lines that should be off. This makes it so the proper value is on the data bus. When no circuits are connected to the data bus all the data lines are on. There are four other lines of the data bus which include ground, positive 5 volts, the clock signal, and clear. When the clear line is connected to ground it sets the counters to 1 and the output register to zero.

10 Bytes of 4-Bit Memory

The memory module consists of the memory display with buffers and 10 bytes of memory. When memory is being sent to the data bus the display will show which byte is being sent. Each byte of memory is hard coded based on whether resistors are connected to positive or ground. If the resistor is connected to the ground it is on and if it is connected to 5 volts it is off. This is because each byte is connected to the data bus via simplified tri-state buffers which invert the value. The first seven bytes of memory store the

opcode instructions, byte 8 contains the load A value, byte 9 is the add value, and byte 10 holds the subtract value.

Accumulator Register

Data is sent to the accumulator register via the data bus. These values get latched in the master-slave data flip-flop and are continuously being sent into the ALU. During the 5 and 6 stages of the ring counter cycle, the value for the ALU will be sent to the data bus and latched into the accumulator register. This makes it so the value the ALU computes can be used in the next ring counter cycle.

Output Register

The output register is made with four regular data flip-flops. When the entire program is complete the final value gets latched into the output register. How this computer is built the output is displayed on LEDs but does not get sent to other circuits.

Register B

Data gets sent from the memory module to the data bus to get latched in register B. Once the value gets latched it is continuously being sent into the ALU. Each bit value gets latched into a regular data flip-flop. This register is built very similarly to register A.

Arithmetic Logic Unit, ALU

The ALU allows the computer to add and subtract numbers up to 15. This is done by using four **full adders** and four **XOR** subtract gates. Data is sent into the ALU from the accumulator register and register B. The output is displayed on LEDs and is sent to the data bus via tri-state buffers.

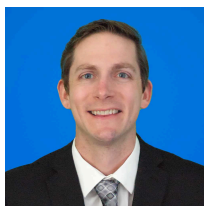
Most people are familiar with the base 10 number system. Computers operate using the base two number system which is also called binary. The video above shows how computers add numbers by using a 4-bit calculator I built on breadboards. If you watch this you will have a good understanding of how the ALU of the computer works.

This video shows how to build a **4-bit calculator** using individual transistors. Four full adders are built to be able to add numbers up to 31 as the output has 5-bits.

Final Thoughts

Building a transistor computer on breadboards helps explain how a computer works. It also helps teach logic gate circuit design, data flip-flop design, and how to build each circuit of the computer. If I tried to build the processor in my current personal computer which has about 3 billion transistors it would take about 100 million breadboards to build. This computer also has 32 GB of RAM which has around 35 billion transistors, the 8GB NVIDIA GPU has around 7 billion transistors, and the 1 TB SSD has around 3 trillion transistors. So clearly this could not be built on breadboards.

However, when computers were first being built they were much larger than this breadboard computer. As technology progressed transistors and computers were able to be built much smaller. My next goal is to build artificial neurons. At first, they will be large and bulky but eventually, we should be able to build billions of neurons and trillions of synapses on a very small silicon or carbon substrate.



Cody Wabiszewski

Cody started the Global Science Network with the idea people should be focusing more time, energy, and resources on useful projects. He has a bachelor's degree in aerospace engineering and a master's degree in mechanical engineering. Cody has worked for the US federal government, a university, a large corporation, small businesses, and for himself. He has done human brain computer interface research and is currently working towards creating non-biological human consciousness.

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